Federated Learning for CIFAR-10 Image Classification under Non-IID Client Partitions

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Abstract

Federated learning trains a shared model across distributed clients without moving the clients' raw data to a central server. This report studies that setting on CIFAR-10 image classification using a ResNet-18 model and non-IID client partitions generated with a Dirichlet concentration parameter of $\alpha = 0.5$. A centralized full-precision baseline is compared against two federated strategies: Federated Averaging (FedAvg) and Federated Distillation/Fusion (FedDF). The centralized model achieves the strongest result, reaching 92.67% test accuracy and 0.2481 test loss. Among the federated runs, FedAvg performs best overall with 78.58% test accuracy, while FedDF reaches 74.72%. The class-level confusion matrices show that non-IID partitioning creates uneven behavior across classes, most notably a collapse of FedAvg recall for the CIFAR-10 cat class despite strong recall on several other classes. These results confirm the central trade-off of federated learning: privacy-preserving distributed training is feasible, but statistical heterogeneity can substantially reduce accuracy and stability relative to centralized training.

1 Introduction

Many machine-learning workflows assume that all training data can be collected in one location. In practice, this assumption can fail because data are private, regulated, too large to transfer, or owned by multiple organizations. Federated learning (FL) addresses this problem by sending model parameters to clients, allowing clients to train locally, and aggregating model updates on a server rather than centralizing the raw data [1, 2].

This project evaluates FL on CIFAR-10 image classification. The main question is how much performance is lost when training is moved from a centralized baseline to federated training with heterogeneous client data. The report focuses on three methods:

- a centralized ResNet-18 baseline trained on the full training split;
- FedAvg, the standard weighted parameter-averaging strategy;
- FedDF, a distillation-based federated fusion strategy using auxiliary unlabeled data.

The experimental results show that the centralized baseline remains the best-performing configuration. However, FedAvg and FedDF both learn useful global models under non-IID partitioning, with FedAvg achieving the higher aggregate test accuracy and FedDF showing a less extreme failure on the cat class.

2 Background and Related Work

2.1 Centralized and Federated Training

In centralized learning, the server has direct access to the complete training data. This simplifies optimization because all batches are sampled from the same global distribution, but it creates privacy, storage, and communication concerns. In federated learning, clients retain their local datasets and exchange model parameters or updates with the server. This reduces raw-data movement, but it introduces new challenges: client data can be non-IID, client devices can have different compute capabilities, and communication can become a bottleneck [2].

2.2 Federated Averaging

FedAvg is the canonical FL algorithm introduced by McMahan et al. [1]. At communication round t, the server broadcasts the global model w_t to selected clients. Each client k performs local training on its private dataset and returns an updated model $w_{t+1}^{(k)}$. The server computes a weighted average,

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{\sum_{j=1}^{K} n_j} w_{t+1}^{(k)}, \tag{1}$$

where n_k is the number of samples on client k. FedAvg is communication-efficient because clients can perform multiple local steps before communicating. Its weakness is that local updates can drift in different directions when client distributions are highly heterogeneous [3, 4].

2.3 Federated Distillation/Fusion

FedDF improves global model fusion by using ensemble distillation rather than simple parameter averaging [5]. In the implementation evaluated here, client models act as an ensemble teacher over an auxiliary unlabeled dataset, and the server trains a global student model to match the teacher distribution. With temperature T, the server minimizes a distillation objective of the form

$$\mathcal{L}_{\mathrm{DF}} = T^2 \, \mathrm{KL}(\bar{p}_T(y \mid x) \,\|\, p_T(y \mid x; w)) \,, \tag{2}$$

where $\bar{p}_T$ is the averaged teacher prediction and p_T is the global student prediction. This can reduce the mismatch caused by averaging parameters from models trained on different client distributions, but it depends on the quality and relevance of the auxiliary data.

3 Methodology

3.1 Task and Dataset

The task is 10-class image classification on CIFAR-10 [6]. CIFAR-10 contains small natural images from the classes airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The experiments use 45,000 training images, 5,000 validation images, and 10,000 test images for the centralized baseline and FedAvg. The FedDF run uses 35,000 client samples, 10,000 server samples, and 10,000 unlabeled STL-10 samples for distillation [7].

The federated runs use three clients. Client partitions are non-IID and generated with a Dirichlet concentration parameter of $\alpha = 0.5$. Lower α values create more class imbalance across clients, so this setting intentionally tests the robustness of each aggregation strategy under heterogeneous data.

3.2 Model

All experiments use a ResNet-18 architecture adapted for CIFAR-10 classification [8]. The centralized baseline trains a single model directly on the full training split. The federated experiments use the same model family but train through server-client rounds. Accuracy and cross-entropy loss are measured on the validation and test splits.

3.3 Federated Implementation

The project follows the Flower framework workflow for server-client federated training [9]. In each federated round, clients train locally, return updates or client models, and the server produces the next global model using the selected strategy. FedAvg aggregates client parameters using client sample counts. FedDF uses server-side distillation with temperature $T = 4$ and an auxiliary unlabeled STL-10 distillation set.

4 Experimental Setup

Table 1 summarizes the three exported runs. The centralized baseline is trained for 30 epochs, while the federated configurations are trained for 10 communication rounds.

The primary metrics are validation accuracy, validation loss, test accuracy, and test loss. Training time is also recorded from the exported history files. For the centralized baseline and FedDF, the reported time is the sum of per-epoch or per-round durations. For FedAvg, the history file records cumulative elapsed time, so the final elapsed time is reported.

Table 1: Experimental configurations from the exported project summaries.

Method	Model	CIFAR-10 training data	Additional data	Partitioning	Length
Centralized	ResNet-18	45,000 train, 5,000 validation	None	Centralized	30 epochs
FedAvg	ResNet-18	45,000 client samples, 5,000 validation	None	3 clients, Dirichlet $\alpha = 0.5$	10 rounds
FedDF	ResNet-18	35,000 client samples, 10,000 server samples, 5,000 validation	10,000 unlabeled STL-10 distillation samples	3 clients, Dirichlet $\alpha = 0.5$	10 rounds

5 Results

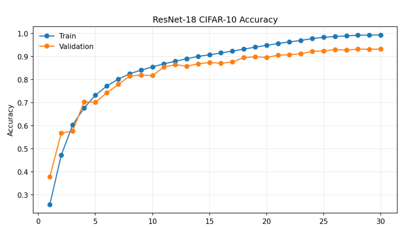
5.1 Aggregate Performance

Table 2 gives the main validation and test results. The centralized baseline achieves the best test accuracy, 92.67%, and the lowest test loss, 0.2481. FedAvg is the stronger federated method, reaching 78.58% test accuracy. FedDF reaches 74.72% test accuracy and peaks at validation round 7, after which validation accuracy decreases.

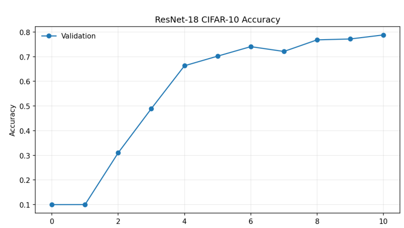
Table 2: Main performance comparison. Accuracy values are percentages.

Method	Val. loss	Val. acc.	Test loss	Test acc.	Time	Best point
Centralized	0.2350	93.22	0.2481	92.67	15.9 min	epoch 30
FedAvg	0.6102	78.80	0.6247	78.58	32.2 min	round 10
FedDF	0.7081	75.04	0.7118	74.72	25.5 min	round 7

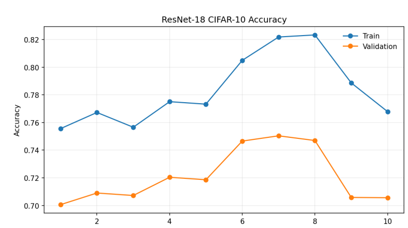
The centralized baseline exceeds FedAvg by 14.09 percentage points and FedDF by 17.95 percentage points in test accuracy. FedAvg exceeds FedDF by 3.86 percentage points. This gap indicates that moving to federated training under non-IID data has a larger effect than the difference between the two federated strategies in this experiment.



(a) Centralized

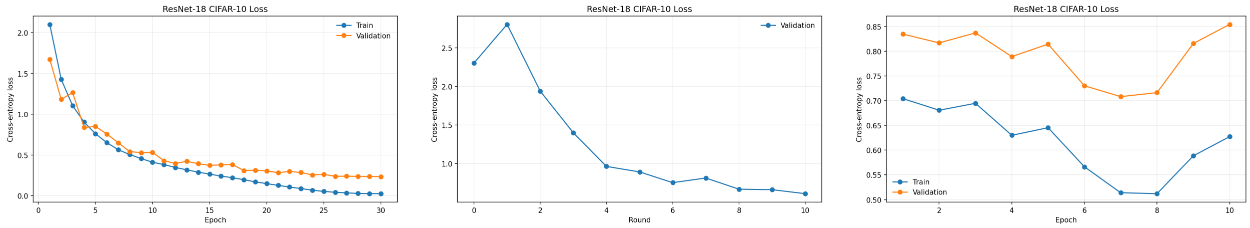


(b) FedAvg



(c) FedDF

Figure 1: Accuracy curves for the centralized baseline and federated strategies. The centralized model converges to the highest validation accuracy, FedAvg improves through round 10, and FedDF reaches its best validation accuracy at round 7.



(a) Centralized (b) FedAvg (c) FedDF

Figure 2: Loss curves for the three runs. The centralized baseline achieves the lowest validation loss. The federated methods have higher loss under the non-IID client partition.

5.2 Class-Level Behavior

Overall accuracy hides important class-level differences. Table 3 reports per-class recall computed from the confusion matrices. The centralized baseline is balanced across classes, with recall above 84% for every class. FedDF is weaker overall but still recognizes every class at non-trivial recall. FedAvg, despite having the best federated test accuracy, almost completely misses the cat class, with only 0.6% recall.

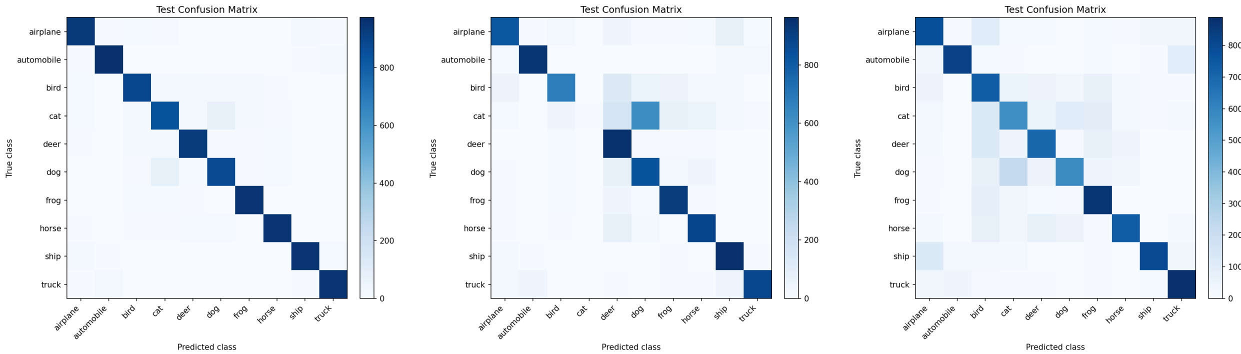
Table 3: Per-class recall from the test confusion matrices. Values are percentages.

Class	Centralized	FedAvg	FedDF
Airplane	93.7	81.7	78.3
Automobile	97.4	93.8	82.9
Bird	89.2	67.5	73.5
Cat	84.6	0.6	56.8
Deer	92.6	95.4	69.9
Dog	87.0	83.7	57.9
Frog	95.8	91.0	86.6
Horse	95.4	88.5	73.4
Ship	95.2	96.2	79.1
Truck	95.8	87.4	88.8

6 Discussion

The results support three main observations. First, centralized training remains the upper bound in this experiment because it can optimize directly over the complete CIFAR-10 training distribution. Its validation and test accuracies are close, suggesting good generalization rather than simple overfitting.

Second, FedAvg is the best federated method by aggregate accuracy. This is expected because FedAvg directly optimizes the target CIFAR-10 task using all 45,000 client samples. However, the confusion matrix shows that aggregate accuracy can hide dangerous failures: the model performs very well on deer, ship, automobile, and frog, but fails on cat. This is consistent with client drift under non-IID partitions, where local updates overrepresent some class mixtures and underrepresent others.



(a) Centralized (b) FedAvg (c) FedDF

Figure 3: Test confusion matrices. FedAvg has strong recall on several classes but severely confuses cats with dog and deer-like classes, illustrating how non-IID client data can produce class-specific failures.

Third, FedDF does not outperform FedAvg in this run, but its class-level behavior is less extreme for cat recognition. The auxiliary STL-10 distillation set may help smooth the global model by exposing the server to broader natural-image structure, but the domain mismatch and reduced number of client samples likely limit the final accuracy. FedDF also peaks before the final round, indicating that model selection by validation accuracy is important for distillation-based strategies.

7 Limitations

This study is limited to one dataset, one model architecture, three clients, one Dirichlet heterogeneity level, and one exported run per method. Because no repeated random seeds are included, the results should be interpreted as a project comparison rather than a statistically complete benchmark. The experiment also evaluates privacy from the perspective of data locality only; it does not implement differential privacy, secure aggregation, or defenses against gradient leakage. Finally, communication cost is represented indirectly through round time rather than explicit byte-level accounting.

8 Conclusion

The project demonstrates the main practical trade-off in federated image classification. A centralized ResNet-18 trained on CIFAR-10 achieves 92.67% test accuracy. Under non-IID federated training with three clients and $\alpha = 0.5$, FedAvg reaches 78.58% and FedDF reaches 74.72%. FedAvg is the strongest federated strategy overall, but its class-level collapse on cats shows that non-IID partitions can create serious hidden errors. Future work should evaluate more clients, repeated seeds, stronger heterogeneity controls, and additional strategies such as FedProx, SCAFFOLD, or adaptive client weighting.

Reproducibility Note

The reported metrics and figures are taken from the supplied project exports in `prism-uploads/`: `summary.csv`, `summary_2.csv`, `summary_3.csv`, `history.csv`, `history_2.csv`, `history_3.csv`, and the corresponding accuracy, loss, and confusion-matrix image files.

References

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